

Experiments continued

What matters for compatibility recovery?

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Recap

Results

- Source-only learned utility family
- Source-only AE family
- Target-local direct support-NELBO

Key takeaways

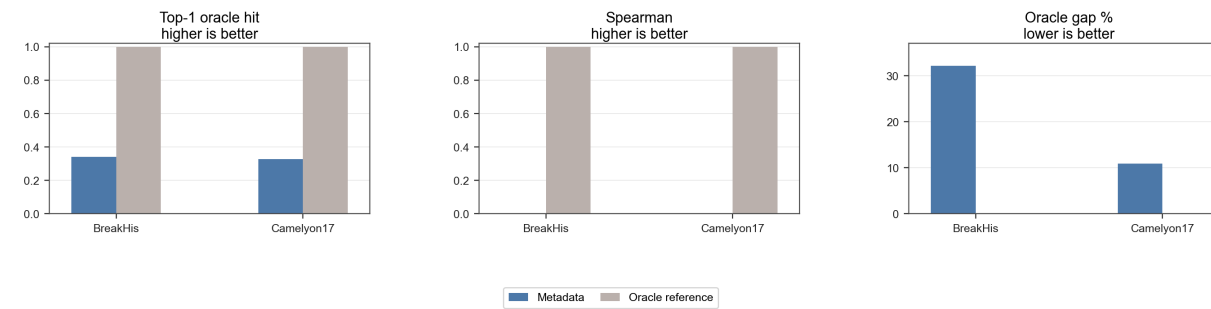
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Recap

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Metadata-Baseline

Metadata similarity does not suffice for compatibility routing



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Where we stopped last time

Key points

- Metadata routing changes expert selection but does not reliably improve utility
- Learned proxy models and metadata-conditioned variants did not recover utility-aligned rankings
- The bottleneck was localized to compatibility estimation, not the routing mechanism itself

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The new question

How can we estimate compatibility without leakage?

- Learned utility predictors
 - Pairwise ranker
- Source-only AE confidence family
 - Raw AE confidence
 - AE utility calibrator
- Target-local direct support-NELBO

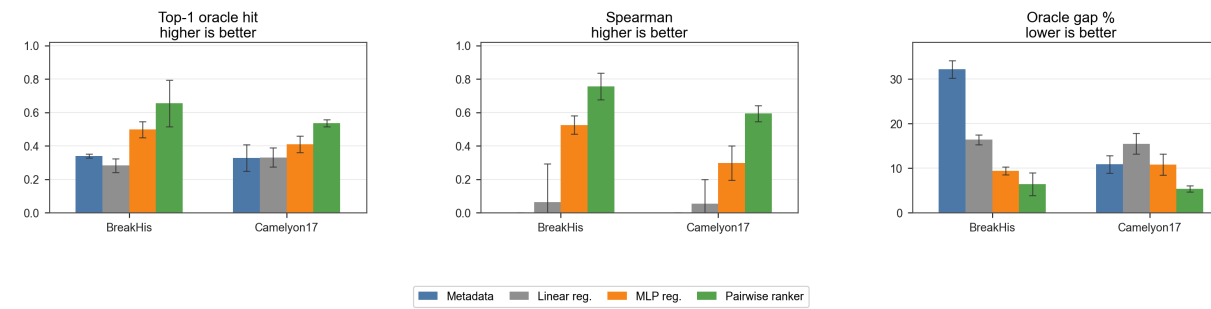
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Results

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Result A

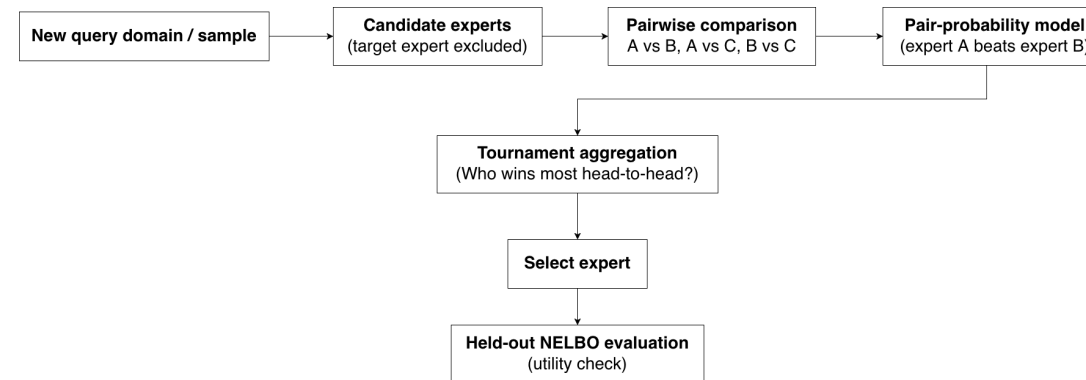
Learned predictors recover signal, but fail adoption gates



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Pairwise probability tournament routing

Estimate who beats whom and select the tournament winner



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The key change is that I stop asking for an absolute utility prediction per expert.

I instead learn pairwise preferences: for this query, is expert A more likely to outperform expert B?

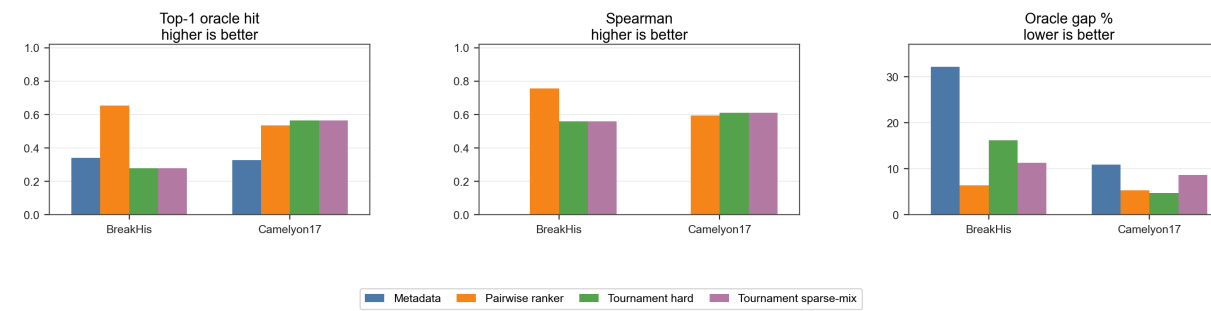
These probabilities form a tournament over all eligible source experts.

The winner becomes the routed expert.

Importantly, the target expert is excluded, and the final claim is judged only by held-out NELBO.

Result B

Pairwise tournament improves ranking, but remains unstable

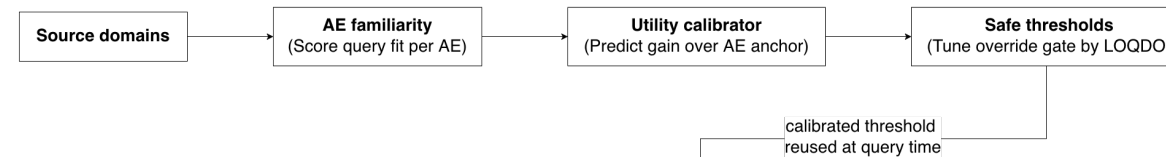


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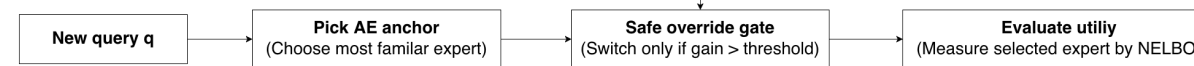
AE Utility calibrated safe override

Source-only AE familiarity becomes a gated routing decision

1. Source-only calibration loop



2. Query-time routing and evaluation



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This method has two phases.

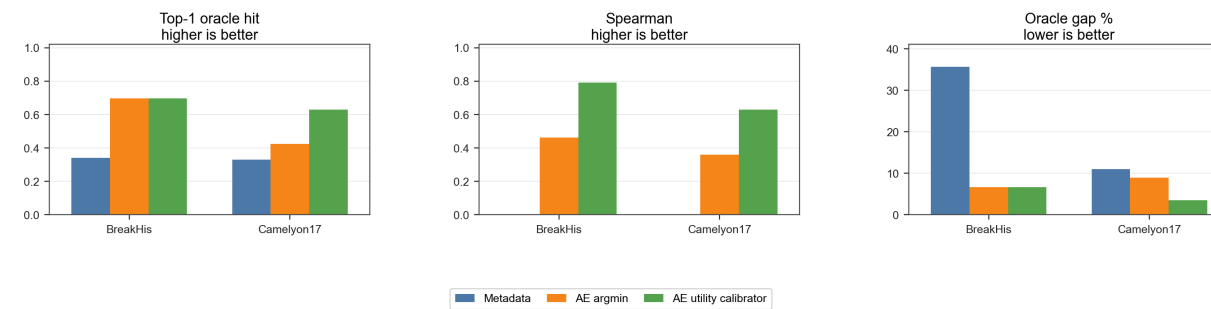
First, source-only calibration learns when AE familiarity is actually predictive of utility.

Then at query time, we start from the AE-best expert but only override it when the calibrated source-inner threshold says the predicted gain is strong enough.

The final claim is not based on AE similarity; it is judged afterward with held-out NELBO utility.

Result C

AE calibration helps, but remains dataset-dependent



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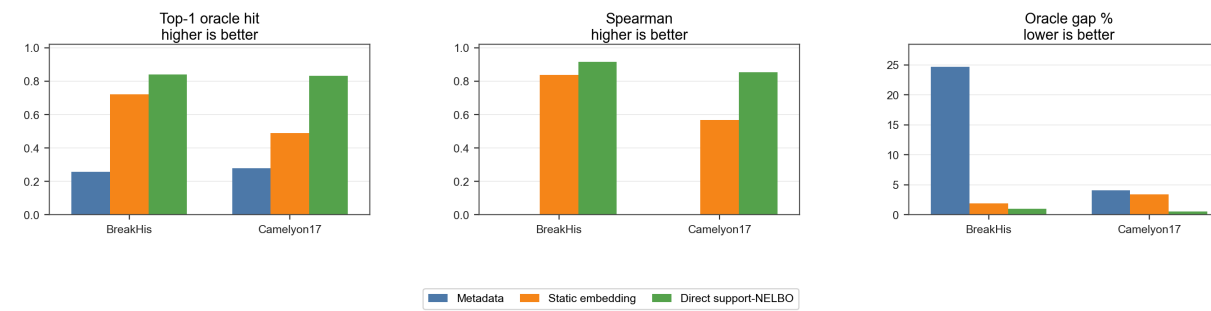
The idea is: each source domain trains its own Autoencoder on frozen feature embeddings.

At test time, a target/query sample is passed through every source AE.

If source AE e reconstructs the query embedding well, the query is assumed to lie near source domain e 's learned feature manifold.

Result D

Target-local NELBO turns compatibility into measurable utility



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Key takeaways

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Key takeaways

Hypothesis	Methods	Outcome
Domain identity is enough	Metadata routing	No
Source-only utility proxies are enough	Learned utility, pairwise tournament, AE confidence, AE calibrator	Useful but unstable
Target-local utility is needed	Direct support-NELBO	Strongest thesis-facing result

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Outlook

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Thank you!

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